C-HI Restart Segment Quick-look

Date: 2026-05-09

Summary

This report records the first restart-continuation segment for the particles-enabled C-HI one-node shakedown run on kunanyi.

Run facts:

- source restart job: 690452.kunanyi-ohpc
- job name: c_hi_restart
- scheduler result: Exit_status = 0
- resources: select=1:ncpus=28:mpiprocs=28
- node: n129
- PBS walltime used: 01:40:09
- PBS CPU time used: 44:14:47
- PBS peak memory: 30588076kb
- template restart mode: RUN_KIND=h5_restart_1
- PLUTO command: -i pluto.ini -e environment.ini -maxtime 1.5 -h5restart 1

The segment restarted from the previous dbl.h5 output and advanced the simulation from 0.4474 Myr to a new restart-capable output at 1.0101 Myr. The run wrote a fresh markers/DONE file and no FAILED marker was observed.

This remains an early run-health diagnostic, not a science result. It is, however, the first full confirmation that the static-grid HDF5 restart path is working for the particles-enabled Stewart-cluster C-HI setup.

Output Catalogue

The current HDF5 catalogue illustrates why output numbering cannot be treated as a simple physical-time sequence once walltime-forced products are present.

Output	Type	Time (Myr)	Step	Role
0	dbl.h5	0.0000	0	initial restart-capable output
1	dbl.h5	0.4474	583	first walltime-forced restart output
2	dbl.h5	1.0101	1146	second walltime-forced restart output
0	flt.h5	0.0000	0	initial float output
1	flt.h5	0.4474	583	first walltime-forced float output

Output	Type	Time (Myr)	Step	Role
2	`flt.h5`	0.9990	1135	nominal 1 Myr science-cadence output
3	`flt.h5`	1.0101	1146	second walltime-forced float output

The corresponding CSV is stored at `docs/science/assets/c_hi_restart_quicklook_2026-05-09/output_catalogue.csv`.

Particle outputs advanced from `particles.0045.dbl` after the first segment to `particles.0103.dbl` after this restart segment.

Quick-look Extraction

A one-core compute-node extraction job inspected the nominal 1 Myr float output:

- quick-look extraction job: `690479.kunanyi-ohpc`
- inspected file: `/scratch/bwbrook/xray_cavities_runs/2026-05-08-c-hi-segment/C-HI/particles/problem/outputs/data.0002.flt.h5`
- timestep: `Timestep_2`
- PLUTO code time: `306.2961`
- physical time: `0.9990 Myr`
- extraction result: `Exit_status = 0`

The compact evidence bundle was retrieved to:

```
build/science/kunanyi_stage_2026-05-08/c_hi_restart_690452_quicklook_690479_compact_2026-05-0
```

SHA-256:

```
fe6c5259dcbe95d759d965d5b2e574e323af168dd127469a9c39d404e2cec2f2
```

Tracer Extent

The default Level 2 truth threshold remains `tracer > 5e-3`. At about 1 Myr, the tracer-defined region is still compact relative to the one-node shakedown domain:

Threshold	Cells	x range (kpc)	y range (kpc)	z range (kpc)
`1e-3`	2572544	-3.9 to 3.9	-3.9 to 3.9	-10.2 to 10.2
`5e-3`	2340392	-3.84 to 3.84	-3.84 to 3.84	-10.1 to 10.1
`1e-2`	2236368	-3.77 to 3.77	-3.77 to 3.77	-9.92 to 9.92

For comparison, the previous `0.4474 Myr` quick-look had a default-threshold extent of about `x,y = +/-2.33 kpc` and `z = +/-6.22 kpc`.

Figures

Full-domain mid-plane maps:

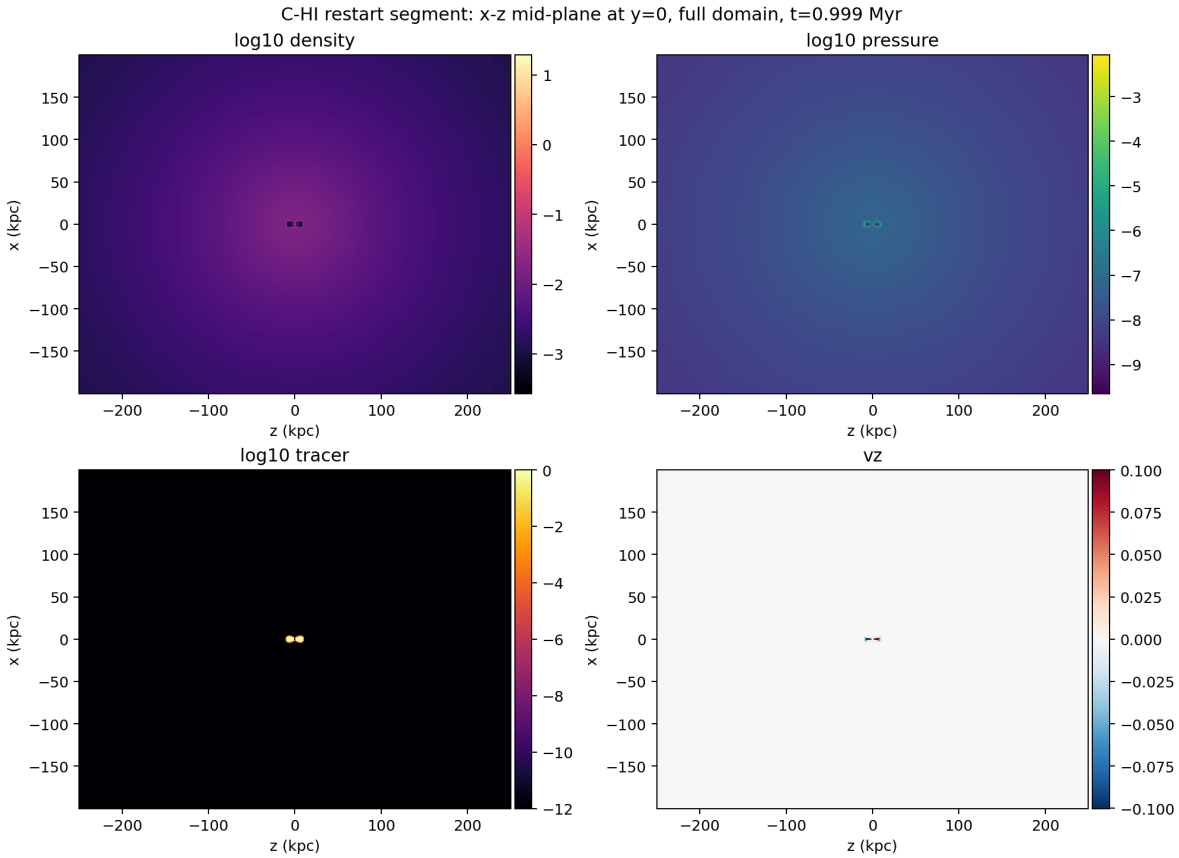


Figure 1: C-HI restart full-domain quick-look maps

Central zoom:

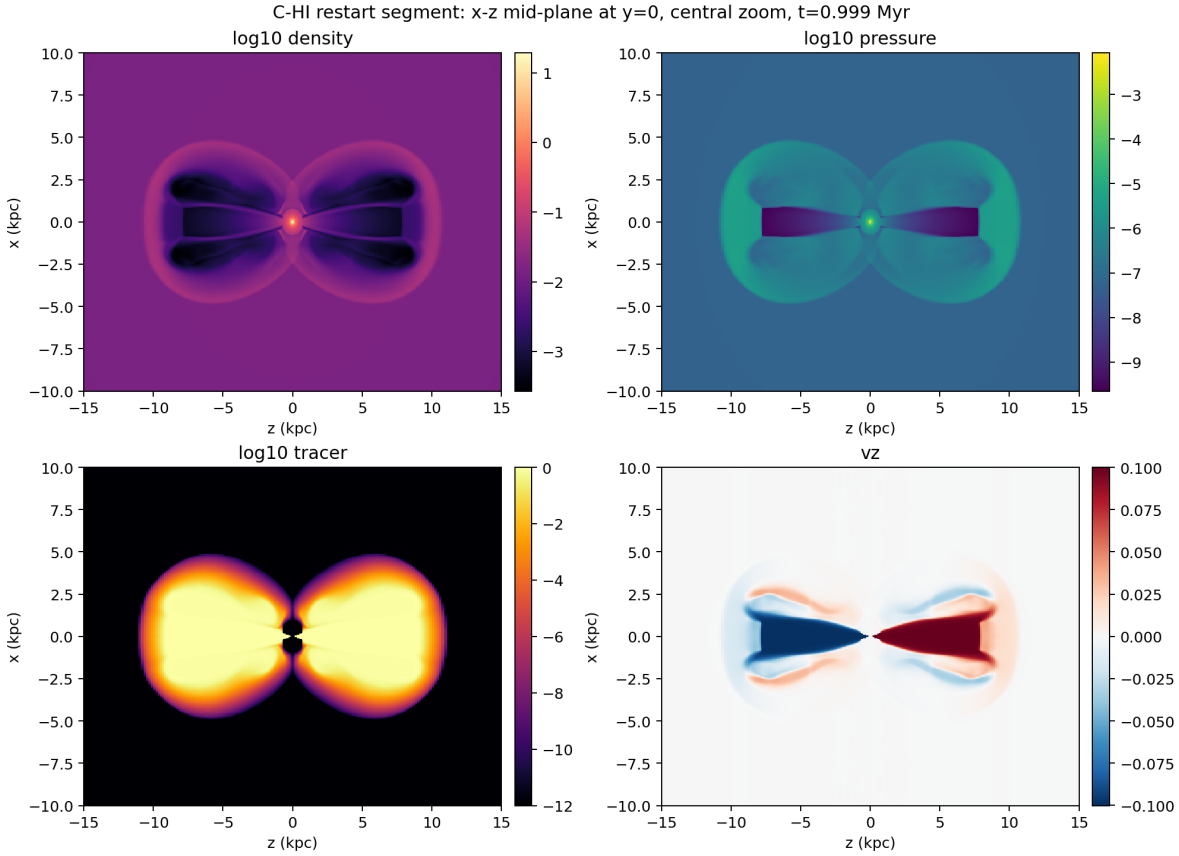


Figure 2: C-HI restart central quick-look maps

Jet-axis profiles:

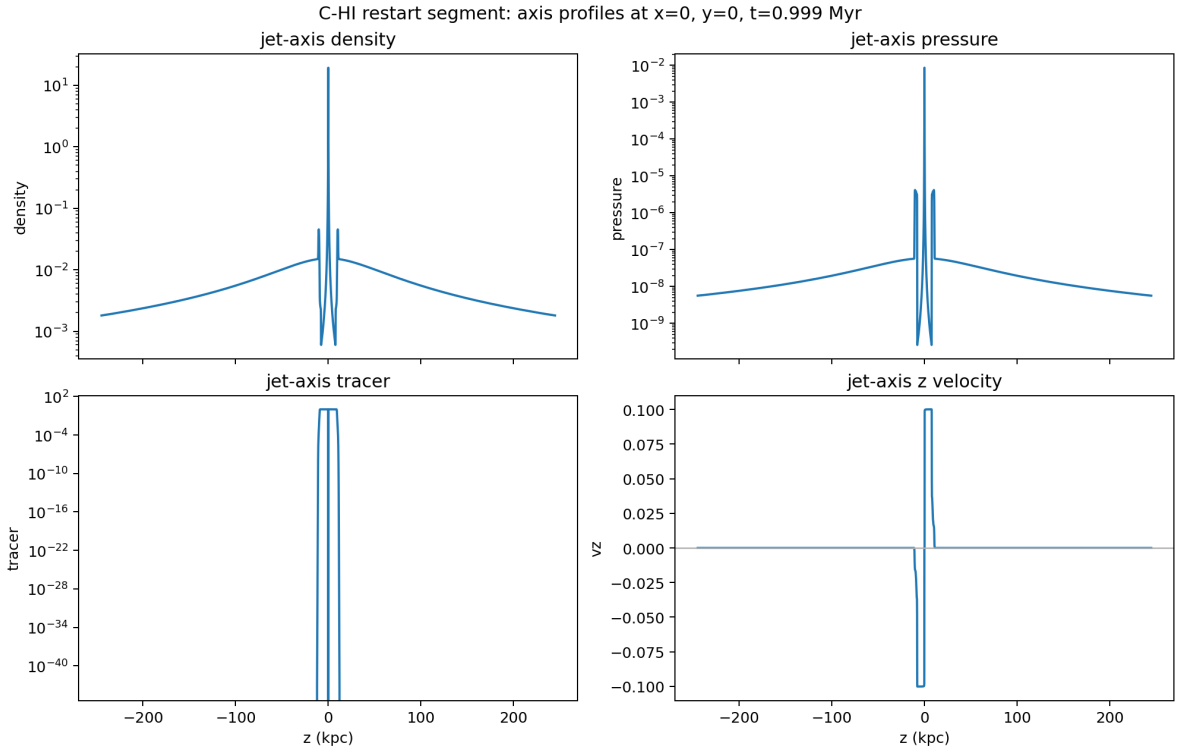


Figure 3: C-HI restart quick-look axis profiles

Interpretation

This segment passes the immediate restart gate:

- the production PBS template detected `data.0001.dbl.h5` and launched PLUTO with `-h5restart 1`
- PLUTO stopped through the requested `-maxtime 1.5` path
- `data.0002.dbl.h5` was written as a new restart-capable output
- the nominal 1 Myr `flt.h5` output was written and extracted successfully
- particle output numbering continued through `particles.0103.dbl`
- tracer, density, pressure, and `v_z` quick-looks remain symmetric and far from the outer grid boundaries

The next useful compute rung is an intermediate multi-node rehearsal, not a full 48-hour production run. That should test the same restartable `C-HI` workflow with a multi-node PBS allocation before promoting to the supervisor grid or a longer early-snapshot sequence.